

Time-Scale-Separated Adaptive Synchronization of Van der Pol Oscillators under Frequently Sampled Couplings

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Abstract—We consider a network of Van der Pol oscillators with heterogeneous, adjustable parameters, where coupling and parameter updates are applied at possibly aperiodic sampling instants. A sequence of coordinate transformations, combining the classical average/disagreement decomposition with a jump-compensated parameter transformation, reveals the different time scales of the closed-loop dynamics and yields a convenient hybrid multi-scale representation. This structure allows us to analyze the reduced dynamics and the boundary-layer dynamics separately using singular-perturbation arguments for hybrid systems. The reduced dynamics comprise an averaged Van der Pol oscillator together with a parameter-disagreement subsystem driven by a persistently exciting regressor, while the boundary layer is governed by a contractive disagreement map. We show that, for sufficiently small initial disagreement and sufficiently frequent communication and suitable coupling and adaptation gains, the oscillator states synchronize asymptotically and the parameter estimates converge to a common constant value.

Index Terms—Heterogeneous networks; Nonlinear oscillators; Adaptation laws; Multi-scale hybrid systems; Asymptotic synchronization.

I. INTRODUCTION

Synchronization of networked dynamical systems is a fundamental problem that arises in a wide range of applications, including cooperative robotic systems [2], electric power networks [3], biological oscillators [4], and distributed sensing and estimation architectures [5]. A substantial portion of the existing literature has focused on networks of *homogeneous* agents, for which all subsystems possess identical intrinsic dynamics. In such settings, the common structure of the individual agents often allows one to establish asymptotic synchronization by means of diffusive couplings and suitably constructed Lyapunov functions [6].

The synchronization problem becomes significantly more challenging when the agents are *heterogeneous*. In the presence of nonidentical dynamics, standard coupling mechanisms are generally unable to eliminate the mismatch among the agents completely. Consequently, one typically obtains only *practical synchronization* [7], [8], [9]. In [10], however, asymptotic output synchronization can be achieved through nonlinear output regulation. A new paradigm to address exact coordination in heterogeneous setting is to endow the agents with the ability to *adapt their dynamics* on the basis of the observed behavior of neighboring agents. Instead of treating heterogeneity as a fixed source of synchronization error, the

idea is to compensate for it online through appropriately designed adaptation mechanisms. This viewpoint was investigated in [11] for continuous-time systems, where adaptation laws were incorporated in network dynamics to reshape the behavior of each agent toward a common synchronized evolution.

In the present work, we consider the case in which the exchange of information among neighboring agents is *time-sampled*. In many applications, communication links are not continuously available, and the agents can only access relative information at isolated sampling instants [12]. As a result, the overall closed-loop system acquires a hybrid nature: the plant states evolve continuously between communication instants, whereas synchronization and adaptation updates occur through jump dynamics. The simultaneous presence of continuous-time dynamics, sampled-data interconnections, and adaptive updates leads to a significantly more intricate analysis than in the continuous-time setting.

Our approach is motivated by the framework introduced in [13], where an analysis method for interconnected hybrid systems was developed based on the decoupling of variables evolving on different time scales. The central idea is to separate fast synchronization variables from slower average and adaptation variables, and to analyze the corresponding reduced order subsystems in a coordinated manner. In this paper, we extend that approach to study the synchronization with hybrid and adaptive laws. The adaptation mechanism introduces additional couplings and qualitative features that are absent in our previous study on hybrid interconnections and we note that the stability analyses for the individual subsystems are completely different. In particular, the analysis of the slow subsystem is carried out using continuous-time persistency of excitation condition associated with the regressor. In the literature, we have seen the use of persistency of excitation for stability analysis of hybrid system in [14]. However, the systems that arise in our study involve rather complex interconnections.

The main contribution of the present study is to show that adaptation fundamentally changes the synchronization properties of heterogeneous networks. Whereas heterogeneous agents without adaptation generally achieve only practical synchronization, the proposed adaptive mechanism makes it possible to recover *asymptotic synchronization* despite the intermittent availability of network information.

II. PROBLEM SETTING

We consider N coupled Van der Pol oscillators, where the i -th agent has state $(\omega_i, \xi_i) \in \mathbb{R}^2$ and unknown parameters

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This note is an extended version of [1]. It contains Appendix I and Appendix II, which were removed from [1] due to space constraints.

$\mu_i > 0$, $\nu_i > 0$. The continuous-time dynamics are

$$\dot{\omega}_i = -\omega_i + \xi_i, \quad (1a)$$

$$\dot{\xi}_i = (1 - \mu_i(\omega_i^2 - 1))(-\omega_i + \xi_i) - \nu_i \omega_i. \quad (1b)$$

The unknown parameters appear only in equation (1b), which can be written in linearly parametrized form as

$$\dot{\xi}_i = \psi(\omega_i, \xi_i)\theta_i + \psi_o(\omega_i, \xi_i), \quad (2)$$

with

$$\psi(\omega_i, \xi_i) := \begin{bmatrix} -(\omega_i^2 - 1)(-\omega_i + \xi_i) \\ -\omega_i \end{bmatrix}^\top \in \mathbb{R}^{1 \times 2}, \quad (3a)$$

$$\theta_i := \text{col}(\mu_i, \nu_i) \in \mathbb{R}^2, \quad (3b)$$

$$\psi_o(\omega_i, \xi_i) := -\omega_i + \xi_i. \quad (3c)$$

The underlying assumption about the coupling is that all the oscillators are connected by an undirected, connected graph $\mathcal{G}$ with node set $\{1, \dots, N\}$. The neighbors of an agent $i \in \{1, \dots, N\}$ are denoted by $\mathcal{N}_i$. Each agent only exchanges the output variable ξ_i with its neighbors at some discrete times, that is, the communication between agents is time-sampled. The parameters θ_i may initially be unknown, but they can be adjusted at communication instants according to an adaptation law. No information about the variables θ_i is shared between the agents. For the sake of simplicity, we assume that all agents exchange information simultaneously, and all agents update their states at the same time.

Our focus in this paper is on studying synchronization for system (1a), (1b), but the treatment can be generalized to the case when agents' dynamics are described by (1a), (2) with generic mappings ψ and ψ_o . In that case, it will be assumed throughout that ψ and ψ_o are twice continuously differentiable.

A. Synchronization and Adaptation Laws

Let the jump times be generated by a scalar timer τ , which evolves as follows

$$\epsilon \dot{\tau} \in [\sigma_1, \sigma_2], \quad \tau \in [0, N_0], \quad (4a)$$

$$\tau^+ = \tau - 1, \quad \tau \in [1, N_0], \quad (4b)$$

with small parameter $\epsilon \in (0, 1]$, constants $\sigma_2 \geq \sigma_1 > 0$, and some positive integer $N_0 \in \mathbb{N}$. When $\tau \in [1, N_0]$, a jump may occur and communication/updates are performed by each agent at that instant.

For the aforementioned setup, our goal is to design update laws for the variables ξ_i , and also an adaptation rule for the unknown parameters θ_i . In particular, at each jump, we update ξ_i and θ_i as

$$\omega_i^+ = \omega_i, \quad (5a)$$

$$\xi_i^+ = \xi_i + \kappa \sum_{j \in \mathcal{N}_i} (\xi_j - \xi_i), \quad (5b)$$

$$\theta_i^+ = \theta_i + \alpha \kappa \psi(\omega_i, \xi_i)^\top \left(\sum_{j \in \mathcal{N}_i} (\xi_j - \xi_i) \right), \quad (5c)$$

where $\kappa > 0$ is the coupling gain, and $\alpha > 0$ is the adaptation gain. Thus, θ_i is updated only at jumps, using a gradient-like step proportional to the local output disagreement.

B. Combined System Dynamics

We stack the variables as

$$\omega := \text{col}(\omega_1, \dots, \omega_N) \in \mathbb{R}^N, \quad (6a)$$

$$\xi := \text{col}(\xi_1, \dots, \xi_N) \in \mathbb{R}^N, \quad (6b)$$

$$\theta := \text{col}(\theta_1, \dots, \theta_N) \in \mathbb{R}^{2N}. \quad (6c)$$

Similarly, we consider the matrix-valued function $\Psi : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}^{N \times 2N}$ and the function $\Psi_o : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}^N$, defined as,

$$\Psi(\omega, \xi) := \text{blockdiag}(\psi(\omega_1, \xi_1), \dots, \psi(\omega_N, \xi_N)), \quad (7a)$$

$$\Psi_o(\omega, \xi) := \text{col}(\psi_o(\omega_1, \xi_1), \dots, \psi_o(\omega_N, \xi_N)). \quad (7b)$$

Using this notation, we can write the overall system compactly in the framework of hybrid systems proposed in [15]. The variables are $(\omega, \xi, \theta, \tau)$, with the flow set $\mathbb{R}^{4N} \times [0, N_0]$. The differential equations over this flow set are:

$$\dot{\omega} = -\omega + \xi, \quad (8a)$$

$$\dot{\xi} = \Psi(\omega, \xi)\theta + \Psi_o(\omega, \xi), \quad (8b)$$

$$\dot{\theta} = 0, \quad (8c)$$

$$\epsilon \dot{\tau} \in [\sigma_1, \sigma_2]. \quad (8d)$$

To write the jump equations in compact form, we let $\mathcal{L}$ be the Laplacian of the graph $\mathcal{G}$ that interconnects the agents. The jump set is given by $\mathbb{R}^{4N} \times [1, N_0]$, and the corresponding jump equations are:

$$\omega^+ = \omega, \quad (9a)$$

$$\xi^+ = \xi - \kappa \mathcal{L} \xi, \quad (9b)$$

$$\theta^+ = \theta - \alpha \kappa \Psi(\omega, \xi)^\top \mathcal{L} \xi, \quad (9c)$$

$$\tau^+ = \tau - 1, \quad \tau \in [1, N_0]. \quad (9d)$$

Problem Statement: We are interested in analyzing the stability of the system (8)–(9). In particular, we study conditions for choosing the parameters ϵ , κ , α such that the states of all the agents $(\omega_i, \xi_i, \theta_i)$, for $i = 1, \dots, N$, synchronize *asymptotically*.

III. MODELING FRAMEWORK: MULTI-SCALE HYBRID SYSTEM

In this section, we consider the framework of hybrid dynamical systems that can be used to analyze system (8)–(9). A distinctive feature of the considered models is that it is seen as an interconnection of two variables evolving at different time scales. The relative difference in the evolution of these variables is characterized by a parameter $\epsilon > 0$.

A. Overall Model

The proposed model comprises two variables $(x, y) \in \mathbb{R}^n \times \mathbb{R}^m$ along with their flow and jump equation. We use a timer τ that essentially determines the jump instants.

The state variables of the overall system are $(x, y, \tau) \in \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}$. We consider the sets $C_x \subset \mathbb{R}^n$ and $C_y \subset \mathbb{R}^m$. The state evolves inside the flow set

$$\mathcal{C} := C_x \times C_y \times [0, N_0],$$

for some positive integer $N_0 \in \mathbb{N}$, and undergoes instantaneous changes inside the jump set

$$\mathcal{D} := C_x \times C_y \times [1, N_0].$$

The evolution of the state variables is described by

$$\begin{cases} \dot{x} = f_x(x, y, \epsilon), \\ \dot{y} = f_y(x, y, \epsilon), \\ \epsilon \dot{\tau} \in [\sigma_1, \sigma_2], \end{cases} \quad \text{for } (x, y, \tau) \in \mathcal{C}, \quad (10a)$$

$$\begin{cases} x^+ = g_x(x, y), \\ y^+ = g_y(x, y), \\ \tau^+ = \tau - 1, \end{cases} \quad \text{for } (x, y, \tau) \in \mathcal{D}. \quad (10b)$$

We assume that the sets $C_x \subset \mathbb{R}^n$ and $C_y \subset \mathbb{R}^m$ are closed and contained in open, connected domains $\mathcal{D}_x \subset \mathbb{R}^n$ and $\mathcal{D}_y \subset \mathbb{R}^m$, respectively. The vector fields $f_x : \mathcal{D}_x \times \mathcal{D}_y \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ and $f_y : \mathcal{D}_x \times \mathcal{D}_y \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^m$ are continuously differentiable in all arguments. For the timer dynamics, the constants σ_1 and σ_2 satisfy $\sigma_2 \geq \sigma_1 > 0$. For the jump maps, we assume that the function $g_y : \mathcal{D}_x \times \mathcal{D}_y \rightarrow \mathbb{R}^m$ is locally Lipschitz continuous.

In the problems considered in this paper, the parameter ϵ in (10) is assumed to be small, which corresponds to frequent jumps. We refer to x as the *slow variable* and y as the *fast variable*. The fast variable y is stabilized mainly by the repeated jump map, whose update frequency grows as $1/\epsilon$. The variable x may also jump; the essential structural requirement is that this jump vanish quadratically with the fast deviation.

B. Auxiliary Dynamical Systems

As stated earlier, we are interested in analyzing the behavior of system (10) when the parameter ϵ is sufficiently small, i.e., $\epsilon \approx 0$. In this regime, the trajectories $(x(\cdot), y(\cdot))$ can be approximated by two auxiliary dynamical systems.

The first is the *boundary-layer system*, defined for a fixed value of $x \in C_x$ as

$$y^+ = g_y(x, y), \quad (11)$$

for $y \in C_y$. A small value of ϵ implies that the timer τ evolves rapidly, leading to many jumps within a short continuous-time interval. Hence, starting from an initial condition $(x(0, 0), y(0, 0))$, while $x(\cdot)$ remains close to $x(0, 0)$, the variable $y(\cdot)$ evolves according to the discrete-time system (11).

To define an equilibrium for the discrete system (11), we assume the existence of a twice continuously differentiable function $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$g_y(x, h(x)) = h(x), \quad \forall x \in C_x. \quad (12)$$

Intuitively, $y(t, j)$ rapidly converges to $h(x(0, 0))$ after several jumps and remains near $h(x(t, j))$ as time evolves. The evolution of $x(t, j)$ can thus be approximated by the *quasi steady-state (QSS) system*:

$$\dot{x} = f_x(x, h(x), 0), \quad x \in C_x, \quad (13)$$

which represents the dynamics on the *slow manifold* $\{(x, y) \in \mathbb{R}^n \times \mathbb{R}^m : y = h(x)\}$. Finally, we assume that $h(x) \in C_y$ for all $x \in C_x$ to ensure well-posedness.

C. Stability Assumptions

We now impose stability assumptions on the boundary-layer and quasi steady-state systems. These properties will be used to establish the stability of the full system (10) for sufficiently small ϵ .

Assumption 1. The quasi steady-state system (13) admits an asymptotically stable compact attractor $\mathcal{A}_x \subset C_x$, and there exists a continuously differentiable function $V_x : \mathcal{D}_x \rightarrow \mathbb{R}$ such that, for all $x \in C_x$ and every $\epsilon \in [0, 1]$,

$$\underline{a}\|x\|_{\mathcal{A}_x}^2 \leq V_x(x) \leq \bar{a}\|x\|_{\mathcal{A}_x}^2, \quad (14a)$$

$$\nabla_x V_x(x) f_x(x, h(x), \epsilon) \leq -a\|x\|_{\mathcal{A}_x}^2, \quad (14b)$$

$$\|\nabla_x V_x(x)\| \leq c\|x\|_{\mathcal{A}_x}, \quad (14c)$$

where $\underline{a}, \bar{a}, a, c > 0$ are positive constants.

Assumption 2. For every compact set $K_x \subset C_x$, the set $\mathcal{A}_y := \{y = h(x) \in C_y \mid x \in K_x\}$ is globally asymptotically stable (GAS) for the system (11). In particular, there exists a continuously differentiable function $V_y : \mathcal{D}_x \times \mathcal{D}_y \rightarrow \mathbb{R}$ satisfying:

$$\underline{b}\|y - h(x)\|^2 \leq V_y(x, y) \leq \bar{b}\|y - h(x)\|^2,$$

$$V_y(x, g_y(x, y)) \leq \gamma V_y(x, y),$$

$$\|\nabla_y V_y(x, y)\| \leq \ell_y \|y - h(x)\|,$$

$$\left\| \nabla_x V_y(x, y) + \nabla_y V_y \frac{\partial h}{\partial x} \right\| \leq \ell_x \|y - h(x)\|^2,$$

for all $(x, y) \in K_x \times C_y$, where $\underline{b}, \bar{b}, \gamma, \ell_y, \ell_x > 0$ are constants and $0 < \gamma < 1$.

In addition to the above stability assumptions on the auxiliary systems, we impose the following condition to ensure that the desired attractor corresponds to a forward invariant set.

Assumption 3. For every $x \in \mathcal{A}_x$ and $\epsilon \in [0, 1]$, it holds that

$$f_y(x, h(x), \epsilon) - \frac{\partial h}{\partial x}(x) f_x(x, h(x), \epsilon) = 0. \quad (15)$$

Remark 1. Assumption 3 ensures that the error $z := y - h(x)$ satisfies $\dot{z} = 0$ whenever $x \in \mathcal{A}_x$ and $z = 0$, for every $\epsilon \in [0, 1]$. It is therefore a compatibility condition for invariance of the graph $y = h(x)$ over $\mathcal{A}_x$ during flows.

The next assumption provides a constraint on jumps in x -dynamics which were not present in [16] and allow us to analyze the adaptation law considered in this paper.

Assumption 4 (Quadratically vanishing jump). For every prescribed compact set $C_x \times C_y$, there exists $\ell_J > 0$ such that

$$\|g_x(x, y) - x\| \leq \ell_J \|y - h(x)\|^2. \quad (16)$$

In particular $g_x(x, h(x)) = x$.

The next result is a modification of [16, Theorem 1], and its proof is developed in Appendix I.

Theorem 1. Suppose that Assumptions 1, 2, and 3 hold. Then, for any compact sets $K_x^0 \subset C_x$ and $K_y^0 \subset C_y$, there exist $\ell_J^* > 0$ and $\epsilon^* > 0$ such that, whenever Assumption 4 holds with $\ell_J \leq \ell_J^*$ and $0 < \epsilon \leq \epsilon^*$, every maximal solution to (10) with $x(0,0) \in K_x^0$, $y(0,0) \in K_y^0$ stays bounded and

$$\limsup_{t+j \rightarrow \infty} \|x(t,j)\|_{A_x} = 0, \quad (17)$$

$$\limsup_{t+j \rightarrow \infty} \|y(t,j) - h(x(t,j))\| = 0. \quad (18)$$

IV. STRUCTURAL DECOUPLING AND JUMP-COMPENSATED COORDINATES

We now return to the hybrid system (8)–(9) and introduce coordinates that expose its structure at different time scales. There are two distinct issues to separate. First, synchronization is naturally described by splitting the network variables into their average component and their disagreement component. Second, although the output disagreement is strongly contracted at each communication instant, the original parameter vector θ is not a slow variable in the sense of Section III and a change of parameter coordinates is proposed to carry out the stability analysis. The transformation below is purely an analysis device; it does not modify the algorithm implemented by the agents.

Let $R \in \mathbb{R}^{N \times (N-1)}$ have orthonormal columns spanning $1_N^\perp$, so that $R^\top R = I_{N-1}$, $R^\top 1_N = 0$, $RR^\top = P_R := I_N - \frac{1}{N} 1_N 1_N^\top$. Since the graph is undirected and connected, $\mathcal{L} 1_N = 0$ and the restriction of $\mathcal{L}$ to $1_N^\perp$ is positive definite. We therefore define

$$\Lambda_{\mathcal{L}} := R^\top \mathcal{L} R, \quad \Lambda := I_{N-1} - \kappa \Lambda_{\mathcal{L}}. \quad (19)$$

The properties of matrix R imply that $\mathcal{L} = R \Lambda_{\mathcal{L}} R^\top$, and $\mathcal{L} R = R \Lambda_{\mathcal{L}}$. In particular, if

$$0 < \kappa < \frac{2}{\lambda_{\max}(\mathcal{L})}, \quad (20)$$

then Λ is Schur.

A. Average and disagreement coordinates for the output

The first transformation is the standard decomposition of ξ into a network average and a disagreement component. Define

$$\chi_o := \frac{1}{N} 1_N^\top \xi \in \mathbb{R}, \quad (21a)$$

$$\tilde{\chi} := R^\top \xi \in \mathbb{R}^{N-1}. \quad (21b)$$

Since the columns of $\frac{1}{\sqrt{N}} 1_N$ together with the columns of R form an orthonormal basis of $\mathbb{R}^N$, we recover

$$\xi = 1_N \chi_o + R \tilde{\chi}. \quad (22)$$

The jump law for ξ has an especially simple form in these coordinates (21). Since $1_N^\top \mathcal{L} = 0$,

$$\chi_o^+ = \chi_o, \quad (23)$$

and, using the structure of $\mathcal{L}$, R , and the definition of $\Lambda_{\mathcal{L}}$,

$$\tilde{\chi}^+ = R^\top (I_N - \kappa \mathcal{L}) \xi = (I_{N-1} - \kappa \Lambda_{\mathcal{L}}) \tilde{\chi} = \Lambda \tilde{\chi}. \quad (24)$$

B. Jump-compensated parameter coordinate

For the coordinate change of the parameter dynamics, it is useful to introduce the full disagreement vector

$$e_\xi := P_R \xi = \xi - 1_N \chi_o = R \tilde{\chi}. \quad (25)$$

Thus $\tilde{\chi} = 0$ is equivalent to output synchronization. Equivalently,

$$e_\xi^+ = R \Lambda \tilde{\chi}, \quad e_\xi^+ - e_\xi = -\kappa \mathcal{L} \xi. \quad (26)$$

The second identity in (26) is the observation on which the parameter transformation below is based.

For the jump rule in (9c), the definitions in (26) yield,

$$\theta^+ - \theta = -\alpha \kappa \Psi(\omega, \xi)^\top \mathcal{L} \xi = \alpha \Psi(\omega, \xi)^\top (e_\xi^+ - e_\xi). \quad (27)$$

Thus the leading parameter update is exactly the regressor-weighted change of the output disagreement. In what follows, we observe that if the regressor Ψ does not change during a jump, the quantity $\theta - \alpha \Psi^\top e_\xi$ is invariant across that jump. This suggests compensating θ by precisely this term.

Motivated by (27), define

$$\eta := \theta - \alpha \Psi(\omega, \xi)^\top e_\xi = \theta - \alpha \Psi(\omega, \xi)^\top R \tilde{\chi} \in \mathbb{R}^{2N}. \quad (28)$$

In component form,

$$\eta_i = \theta_i - \alpha \psi(\omega_i, \xi_i)^\top (\xi_i - \chi_o), \quad i = 1, \dots, N. \quad (29)$$

The transformation is invertible once (ω, ξ) are fixed, with

$$\theta = \eta + \alpha \Psi(\omega, \xi)^\top R \tilde{\chi}. \quad (30)$$

Notice also that $\eta = \theta$ whenever $\tilde{\chi} = 0$. Consequently, the new coordinate does not alter the parameter vector on the synchronization manifold; it only removes the part of its transient motion that is tied directly to an instantaneous correction of the output disagreement.

We next compute its jump exactly. Let

$$\Psi := \Psi(\omega, \xi), \quad \Psi^+ := \Psi(\omega^+, \xi^+) = \Psi(\omega, \xi^+),$$

where the second equality follows from $\omega^+ = \omega$. Using (27),

$$\begin{aligned} \eta^+ - \eta &= (\theta^+ - \theta) - \alpha (\Psi^+)^T e_\xi^+ + \alpha \Psi^T e_\xi \\ &= \alpha \Psi^T (e_\xi^+ - e_\xi) - \alpha (\Psi^+)^T e_\xi^+ + \alpha \Psi^T e_\xi \\ &= \alpha [\Psi^T - (\Psi^+)^T] e_\xi^+. \end{aligned} \quad (31)$$

Therefore, by (26),

$$\eta^+ = \eta + \alpha [\Psi(\omega, \xi)^\top - \Psi(\omega, \xi^+)^\top] R \Lambda \tilde{\chi}. \quad (32)$$

Equation (32) is the essential payoff of the transformation. The term on the right-hand side is a product of two quantities that both vanish with the disagreement: the change of the regressor across the jump and the post-jump output disagreement.

The compensation simplifies the jump map, but it does not remove the adaptation mechanism. Rather, that mechanism is transferred to the continuous-time evolution of η . To see this explicitly, use (30) in the flow equation for ξ and define

$$q(\omega, \xi, \eta, \tilde{\chi}) := \Psi(\omega, \xi) \left[\eta + \alpha \Psi(\omega, \xi)^\top R \tilde{\chi} \right] + \Psi_o(\omega, \xi). \quad (33)$$

Then simply $\dot{\xi} = q$. The flow equations become

$$\dot{\omega} = -\omega + 1_N \chi_o + R \tilde{\chi}, \quad (34a)$$

$$\dot{\chi}_o = \frac{1}{N} 1_N^\top q, \quad (34b)$$

$$\dot{\tilde{\chi}} = R^\top q, \quad (34c)$$

$$\dot{\eta} = -\alpha [\dot{\Psi}(\omega, \xi)^\top R \tilde{\chi} + \Psi(\omega, \xi)^\top P_R q]. \quad (34d)$$

Here $\dot{\Psi}$ denotes the derivative of Ψ along the flow, namely

$$\dot{\Psi} = D_\omega \Psi(\omega, \xi)[- \omega + \xi] + D_\xi \Psi(\omega, \xi)[q]. \quad (35)$$

The leading adaptive action has thus moved from the jumps into the continuous-time dynamics of η . Finally, write

$$\begin{aligned} \eta_o &:= \frac{1}{N} (1_N^\top \otimes I_2) \eta, & \tilde{\eta} &:= (R^\top \otimes I_2) \eta, \\ \eta &= (1_N \otimes I_2) \eta_o + (R \otimes I_2) \tilde{\eta}. \end{aligned} \quad (36)$$

The natural partition, consistent with Section III, is therefore

$$x := \text{col}(\omega, \chi_o, \eta), \quad y := \tilde{\chi}, \quad h(x) \equiv 0. \quad (37)$$

V. STABILITY ANALYSIS

We now verify the hypotheses of Theorem 1 for the partition (37). The compensated coordinate makes the verification particularly transparent: the fast subsystem is the disagreement $\tilde{\chi}$, while the effective adaptation appears in the QSS flow of η . Technical derivations are collected in Appendix II.

A. QSS dynamics

On $y = \tilde{\chi} = 0$ one has $\xi = 1_N \chi_o$ and $\theta = \eta$. Define

$$\begin{aligned} \Psi_c(\omega, \chi_o) &:= \Psi(\omega, 1_N \chi_o), \\ \Psi_{o,c}(\omega, \chi_o) &:= \Psi_o(\omega, 1_N \chi_o). \end{aligned} \quad (38)$$

and $q_c := \Psi_c \eta + \Psi_{o,c}$. The QSS system is

$$\dot{\omega} = -\omega + 1_N \chi_o, \quad (39a)$$

$$\dot{\chi}_o = \frac{1}{N} 1_N^\top q_c, \quad (39b)$$

$$\dot{\eta} = -\alpha \Psi_c^\top P_R q_c. \quad (39c)$$

Thus the adaptive law, although implemented at communication instants in the original variables, becomes a continuous-time mechanism in the reduced coordinates.

To expose the QSS structure, set

$$\omega_o := \frac{1}{N} 1_N^\top \omega, \quad \tilde{\omega} := R^\top \omega, \quad (40)$$

and use (36). With $\bar{\psi} := \psi(\omega_o, \chi_o)$, rewriting (39) in averaged and disagreement coordinates yields

$$\dot{\omega}_o = -\omega_o + \chi_o, \quad \dot{\tilde{\omega}} = -\tilde{\omega}, \quad (41a)$$

$$\dot{\chi}_o = \bar{\psi} \eta_o + \psi_o(\omega_o, \chi_o) + d_o, \quad (41b)$$

$$\dot{\eta}_o = d_{\eta_o}, \quad (41c)$$

$$\dot{\tilde{\eta}} = -\alpha (I_{N-1} \otimes \bar{\psi}^\top \bar{\psi}) \tilde{\eta} + d_{\eta_e}, \quad (41d)$$

where, on every fixed compact region, there exists $c_d > 0$ such that

$$|d_o| \leq c_d \|\tilde{\omega}\|, \quad \|d_{\eta_o}\| + \|d_{\eta_e}\| \leq c_d \alpha \|\tilde{\omega}\|. \quad (42)$$

The explicit expressions and bounds are derived in Appendix II-A. In particular, the perturbations vanish when $\tilde{\omega} = 0$. The limiting dynamics then consist of a blended Van der Pol oscillator with constant η_o and a PE-driven gradient system for $\tilde{\eta}$.

For $\theta_o = \text{col}(\mu_o, \nu_o) \in (0, \infty)^2$, consider the second-order system associated with (ω_o, χ_o) dynamics:

$$\dot{s}_\omega = -s_\omega + s_o, \quad (43a)$$

$$\dot{s}_o = \psi(s_\omega, s_o) \theta_o + \psi_o(s_\omega, s_o). \quad (43b)$$

Proposition 1 (Blended dynamics and excitation). For every compact $\Theta_o^* \subset (0, \infty)^2$, (43) has, for each $\theta_o \in \Theta_o^*$, a unique nontrivial hyperbolic orbitally stable periodic orbit $\mathcal{A}_s(\theta_o)$; the family is uniformly locally exponentially orbitally stable. Moreover, there exist $T_{\text{PE}} > 0$ and $0 < \underline{\mu} \leq \bar{\mu}$ such that, uniformly in $\theta_o \in \Theta_o^*$ and along every trajectory on $\mathcal{A}_s(\theta_o)$,

$$\underline{\mu} I_2 \preceq \int_t^{t+T_{\text{PE}}} \psi^\top \psi ds \preceq \bar{\mu} I_2. \quad (44)$$

The proof, including the reduction to the classical Van der Pol equation and a direct verification of (44), is given in Appendix II-B. In what follows, we denote the basin of attraction for $\mathcal{A}_s(\eta_o)$ by $\mathcal{B}(\mathcal{A}_s(\eta_o))$. For a fixed compact set $\Theta_o^* \subset (0, \infty)^2$, the attractor for x -dynamics is defined as,

$$\mathcal{A}_x := \left\{ (1_N s_\omega, s_o, 1_N \otimes \eta_o) : \eta_o \in \Theta_o^*, (s_\omega, s_o) \in \mathcal{A}_s(\eta_o) \right\}. \quad (45)$$

Proposition 2 (QSS stability). Fix compact sets $\Theta_o^* \subset (0, \infty)^2$, $\mathcal{P} \subset \{(s, \vartheta) \in \mathbb{R}^2 \times \text{int } \Theta_o^* : s \in \mathcal{B}(\mathcal{A}_s(\vartheta))\}$, and $R_\eta > 0$ and let $\mathcal{A}_x$ be defined by (45). Then there exist $\alpha_0 \in (0, 1]$, $\bar{r}_\omega, \bar{v} > 0$, a compact set $C_x \subset \{x : \eta_o \in \Theta_o^*\}$ containing $\mathcal{A}_x$, and an open connected $\mathcal{D}_x \supset C_x$, such that,

$$\mathcal{S} := \{x : (s, \eta_o) \in \mathcal{P}, \|\tilde{\omega}\| \leq \bar{r}_\omega, \|\tilde{\eta}\| \leq R_\eta\},$$

one has $\mathcal{S} \subset \text{int } C_x$, and for every $0 < \alpha \leq \alpha_0$, there exists $V_x^\alpha \in C^1(\mathcal{D}_x; \mathbb{R}_{\geq 0})$ satisfying on C_x

$$\underline{a}_x \|x\|_{\mathcal{A}_x}^2 \leq V_x^\alpha(x) \leq \bar{a}_x \|x\|_{\mathcal{A}_x}^2, \quad (46a)$$

$$\nabla V_x^\alpha(x) f_x(x, 0, \epsilon) \leq -\underline{a}_x \alpha \|x\|_{\mathcal{A}_x}^2, \quad \epsilon \in [0, 1], \quad (46b)$$

$$\|\nabla V_x^\alpha(x)\| \leq \hat{c}_x \|x\|_{\mathcal{A}_x}, \quad (46c)$$

where the positive constants $\underline{a}_x, \bar{a}_x, \hat{c}_x$ are independent of α . Moreover, $e^{N_0} \sup_{0 < \alpha \leq \alpha_0} \max_{x \in \mathcal{S}} V_x^\alpha(x) + 1 < \bar{v}$, and, for every compact $\Theta' \subset \text{int } \Theta_o^*$,

$$\{x \in C_x : \eta_o \in \Theta', V_x^\alpha(x) \leq \bar{v}\} \subset \text{int } C_x.$$

The proof is given in Appendix II-C. The constants in (46) and the level $\bar{v}$ are independent of α , and this uniformity is needed for the subsequent application of Theorem 1.

B. Boundary-layer subsystem and synchronization

The remaining hypotheses follow directly from the transformed jump map. The next proposition is obtained from the frozen boundary layer system $y^+ = \Lambda y$, while noting that $R^\top q_c = 0$ on $\mathcal{A}_x$. Further details and proof are given in Appendix II-D.

Proposition 3 (Fast and jump conditions). If (20) holds, then Assumptions 2, 3, and 4 hold for (37). In particular,

$$V_y(x, y) = \|y\|^2, \quad \gamma = \|\Lambda\|^2 < 1, \quad (47)$$

and (16) holds with $\ell_J = c_\eta \alpha \kappa$ for some $c_\eta > 0$.

We can now apply Theorem 1 on any fixed compact region. Fix κ satisfying (20). Proposition 2 provides $\alpha_0 > 0$ with Lyapunov constants uniform for $0 < \alpha \leq \alpha_0$; hence the corresponding threshold $\ell_J^* > 0$ of Theorem 1 can also be chosen uniformly on this interval. Therefore, the slow-jump requirement is satisfied whenever

$$0 < \alpha \leq \alpha_{\text{sp}}^*(\kappa), \quad \alpha_{\text{sp}}^*(\kappa) := \min \left\{ \alpha_0, \frac{\ell_J^*}{c_\eta \kappa} \right\}. \quad (48)$$

The following theorem establishes the main convergence result. Its proof applies the estimates of Theorem 1 on the region furnished by Proposition 2, after an entrance argument, and establishes forward confinement; see Appendix II-E.

Theorem 2 (Adaptive synchronization). Suppose that κ satisfies (20). Let $\mathcal{K}_0 \subset \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^{2N} \times [0, N_0]$ be a compact set of initial conditions for (8)–(9) such that every $(\omega, \xi, \theta, \tau) \in \mathcal{K}_0$ satisfies

$$\theta_o := \frac{1}{N} (1_N^\top \otimes I_2) \theta \in (0, \infty)^2, \quad (\omega_o, \chi_o) \neq (0, 0).$$

Then there exist $\bar{r}_\omega > 0$ and $\alpha^* > 0$ such that, for every $0 < \alpha \leq \alpha^*$, there exists $\epsilon^* > 0$ such that, for every $0 < \epsilon \leq \epsilon^*$ and every admissible aperiodic sampling sequence, every maximal solution issued from $\mathcal{K}_0 \cap \{\|R^\top \omega\| \leq \bar{r}_\omega\}$ is complete and bounded and satisfies, as $t + j \rightarrow \infty$,

$$\begin{aligned} \omega_i(t, j) - \omega_k(t, j) &\rightarrow 0, & \xi_i(t, j) - \xi_k(t, j) &\rightarrow 0, \\ \theta_i(t, j) &\rightarrow \theta_\infty, & \|(\omega_i, \xi_i)(t, j)\|_{\mathcal{A}_s(\theta_\infty)} &\rightarrow 0, \end{aligned} \quad (49)$$

for all $i, k \in \{1, \dots, N\}$, where $\theta_\infty \in (0, \infty)^2$ may depend on the solution.

Remark 2. The hypotheses of Theorem 2 impose no smallness condition on the initial output disagreement $R^\top \xi$ or parameter disagreement $(R^\top \otimes I_2) \theta$. Moreover, for every positive parameter vector, the basin of attraction of the nontrivial periodic orbit of the blended Van der Pol dynamics is $\mathbb{R}^2 \setminus \{0\}$. Hence, the condition $(\omega_o, \chi_o) \neq (0, 0)$ simply excludes the unstable equilibrium of the blended dynamics. The only smallness requirement on the initial state concerns the oscillator disagreement $R^\top \omega$.

Example 1. We illustrate the hybrid adaptive synchronization mechanism on a network of heterogeneous Van der Pol oscillators governed by the equations (8)–(9). The numerical results in Figure 1 confirm the qualitative behavior predicted by the analysis. The outputs ξ_i converge to each other, the internal variables ω_i converge to a common trajectory, and the original parameter disagreement $\tilde{\vartheta} = (R^\top \otimes I_2) \theta$ tends to zero. Thus the adaptive sampled-data coupling eliminates the synchronization error despite heterogeneity in the individual oscillator parameters.

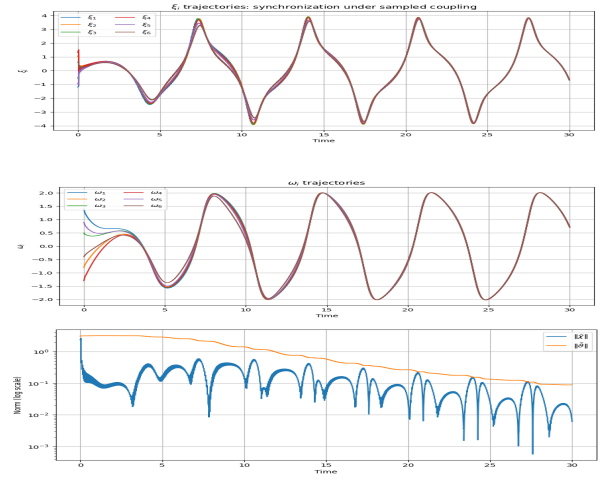


Fig. 1: Plots for (ξ, ω) and the disagreements $(\tilde{\chi}, \tilde{\vartheta})$.

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APPENDIX I PROOF OF THEOREM 1

We follow the proof of the corresponding result in [16] as closely as possible. The main additional point is that the slow variable is now allowed to jump, with a displacement that is quadratic in the distance from the slow manifold. We first fix the compact sets of initial conditions appearing in Theorem 1, and only afterwards choose the admissible upper bounds on ℓ_J and ϵ .

Let $K_x^0 \subset C_x$ and $K_y^0 \subset C_y$ denote these fixed compact sets. Introduce the change of coordinates

$$z := y - h(x)$$

and the closed set $C_X := \{(x, y - h(x)) : (x, y) \in C_x \times C_y\}$. Define

$$V_z(x, z) := V_y(x, z + h(x)).$$

Then, by Assumption 2,

$$\begin{aligned} b\|z\|^2 &\leq V_z(x, z) \leq \bar{b}\|z\|^2, \\ \|\nabla_z V_z(x, z)\| &\leq \ell_y\|z\|, \quad \|\nabla_x V_z(x, z)\| \leq \ell_x\|z\|^2. \end{aligned} \quad (50)$$

The last inequality follows from $\nabla_x V_z = \nabla_x V_y + \nabla_y V_y \frac{\partial h}{\partial x}$.

With $y = z + h(x)$, the flow dynamics in (10) become

$$\begin{aligned} \dot{x} &= f_x(x, h(x), \epsilon) + \hat{f}_x(x, z, \epsilon), \\ \dot{z} &= f_y(x, z + h(x), \epsilon) - \frac{\partial h}{\partial x}(x) f_x(x, z + h(x), \epsilon) \\ &=: f_z(x, z, \epsilon), \\ \epsilon \dot{\tau} &\in [\sigma_1, \sigma_2], \end{aligned} \quad (51)$$

where

$$\hat{f}_x(x, z, \epsilon) := f_x(x, z + h(x), \epsilon) - f_x(x, h(x), \epsilon). \quad (52)$$

At a jump, the transformed jump dynamics are

$$\begin{aligned} x^+ &= g_x(x, z + h(x)), \\ z^+ &= g_y(x, z + h(x)) - h(g_x(x, z + h(x))), \\ \tau^+ &= \tau - 1. \end{aligned} \quad (53)$$

By Assumption 4,

$$\|g_x(x, z + h(x)) - x\| \leq \ell_J\|z\|^2. \quad (54)$$

Choice of a compact region

Choose $\mu > 0$ such that

$$\max_{x \in K_x^0} V_x(x) < \mu, \quad (55)$$

and define

$$\hat{K}_x := \{x \in C_x : V_x(x) \leq e^{N_0}\mu + 1\}. \quad (56)$$

The set $\hat{K}_x$ is compact, since $\mathcal{A}_x$ is compact and the lower bound in (14) makes the sublevel sets of V_x bounded relative to $\mathcal{A}_x$. We now fix the function V_y furnished by

Assumption 2 for this particular set $\hat{K}_x$. Hence the constants $\bar{b}, \bar{b}, \gamma, \ell_y, \ell_x$ are fixed from this point onward. Next choose $\nu > 0$ such that

$$\max_{(x, y) \in K_x^0 \times K_y^0} V_y(x, y) < \nu. \quad (57)$$

Choose constants $c_x, c_y > 0$ satisfying

$$\begin{aligned} 0 &< c_y < \min\{1, \ln(1/\gamma)\}, \\ 0 &< c_x < \min\left\{1, \frac{1}{N_0}, \frac{a}{2\sigma_2\bar{a}}\right\}. \end{aligned} \quad (58)$$

For any $\epsilon \in (0, 1]$, define

$$W_x(x, \tau) := e^{c_x\epsilon\tau} V_x(x), \quad W_z(x, z, \tau) := e^{-c_y\tau} V_z(x, z), \quad (59)$$

and

$$W(x, z, \tau) := \frac{e^{N_0}\mu W_x(x, \tau)}{1 + e^{N_0}\mu - W_x(x, \tau)} + \frac{\nu W_z(x, z, \tau)}{1 + \nu - W_z(x, z, \tau)}. \quad (60)$$

Let $L_0 := e^{2N_0}\mu^2 + \nu^2 + 1$, and define

$$\begin{aligned} \Omega_0^\epsilon &:= \{(x, z, \tau) \in C_X \times [0, N_0] : W_x < 1 + e^{N_0}\mu, \\ &\quad W_z < 1 + \nu, W(x, z, \tau) \leq L_0\}. \end{aligned} \quad (61)$$

Define the transformed set of initial conditions

$$K_0^z := \{(x, y - h(x), \tau) : x \in K_x^0, y \in K_y^0, \tau \in [0, N_0]\}.$$

The choice in (55)–(57) implies, uniformly for every $\epsilon \in (0, 1]$, that

$$K_0^z \subset \text{int } \Omega_0^\epsilon. \quad (62)$$

Indeed, on K_0^z we have $W_x \leq e^{N_0}\mu$ and $W_z \leq \nu$, and hence $W \leq e^{2N_0}\mu^2 + \nu^2 < L_0$.

As in [16], the definition of W also gives uniform upper bounds on its two arguments. More precisely,

$$W_x \leq \frac{(1 + e^{N_0}\mu)L_0}{e^{N_0}\mu + L_0} < 1 + e^{N_0}\mu, \quad W_z \leq \frac{(1 + \nu)L_0}{\nu + L_0} < 1 + \nu \quad (63)$$

throughout Ω_0^ϵ . Consequently,

$$V_x(x) < e^{N_0}\mu + 1, \quad V_y(x, z + h(x)) < e^{N_0}(\nu + 1),$$

where we used $c_y \leq 1$. Therefore, for every $\epsilon \in (0, 1]$, Ω_0^ϵ is contained in the fixed compact set

$$\begin{aligned} \mathcal{K} &:= \{(x, z, \tau) \in C_X \times [0, N_0] : V_x(x) \leq e^{N_0}\mu + 1, \\ &\quad V_z(x, z) \leq e^{N_0}(\nu + 1)\}. \end{aligned} \quad (64)$$

Importantly, $\mathcal{K}$ depends only on the fixed initial compact sets through the choices of μ and ν ; it does not depend on ϵ or ℓ_J .

All constants used below are taken on $\mathcal{K}$ (or on a fixed compact neighborhood of it in $\mathcal{D}_x \times \mathcal{D}_y$) and are therefore independent of $\epsilon \in (0, 1]$ and of ℓ_J in a sufficiently small fixed interval. In particular, from (52),

$$\hat{f}_x(x, z, \epsilon) = \int_0^1 \frac{\partial f_x}{\partial y}(x, h(x) + sz, \epsilon) z ds.$$

Continuous differentiability of f_x therefore gives a constant $N_x > 0$ such that

$$\|\hat{f}_x(x, z, \epsilon)\| \leq N_x \|z\|. \quad (65)$$

We next derive the estimate on f_z directly from the modified Assumption 3. For each x in the projection of $\mathcal{K}$, choose $\bar{x} \in \mathcal{A}_x$ such that $\|x - \bar{x}\| = \|x\|_{\mathcal{A}_x}$. Assumption 3 gives $f_z(\bar{x}, 0, \epsilon) = 0$ for every $\epsilon \in [0, 1]$. Thus

$$\begin{aligned} \|f_z(x, z, \epsilon)\| &\leq \|f_z(x, z, \epsilon) - f_z(x, 0, \epsilon)\| \\ &\quad + \|f_z(x, 0, \epsilon) - f_z(\bar{x}, 0, \epsilon)\|. \end{aligned}$$

Since f_z is locally Lipschitz on the fixed compact region, uniformly for $\epsilon \in [0, 1]$, there exist constants $N_{zz}, N_{zx} > 0$ such that

$$\|f_z(x, z, \epsilon)\| \leq N_{zz}\|z\| + N_{zx}\|x\|_{\mathcal{A}_x}, \quad \epsilon \in [0, 1]. \quad (66)$$

This is exactly the mean-value argument used in [16]. It only requires local Lipschitz continuity of $\frac{\partial h}{\partial x}$ on compact sets; in particular, it is guaranteed if h is twice continuously differentiable, as assumed in [16].

Finally, let $M_x > 0$ be such that

$$\|f_x(x, z + h(x), \epsilon)\| \leq M_x, \quad (x, z, \tau) \in \mathcal{K}, \quad \epsilon \in [0, 1]. \quad (67)$$

For later use, denote by

$$\phi_\mu(s) := \frac{e^{N_0 \mu s}}{1 + e^{N_0 \mu} - s}, \quad \phi_\nu(s) := \frac{\nu s}{1 + \nu - s}.$$

The strict margins in (63) allow us to fix compact intervals, strictly inside the domains of ϕ_μ and ϕ_ν , which contain the corresponding values of W_x and W_z on Ω_0^ϵ . Hence there exist constants

$$0 < m_\mu \leq \phi'_\mu(s) \leq M_\mu, \quad 0 < m_\nu \leq \phi'_\nu(s) \leq M_\nu, \quad (68)$$

on these intervals. The constants in (68) depend on K_x^0, K_y^0 through μ, ν , but not on ϵ or ℓ_J .

Decrease along flows

From Assumption 1, (65), and (58),

$$\begin{aligned} \dot{W}_x &= e^{c_x \epsilon \tau} \left(\epsilon \dot{\tau} c_x V_x + \nabla_x V_x f_x(x, h(x), \epsilon) + \nabla_x V_x \hat{f}_x \right) \\ &\leq e^{c_x \epsilon \tau} \left[- (a - c_x \sigma_2 \bar{a}) \|x\|_{\mathcal{A}_x}^2 + c N_x \|x\|_{\mathcal{A}_x} \|z\| \right] \\ &\leq -\frac{a}{2} \|x\|_{\mathcal{A}_x}^2 + e c N_x \|x\|_{\mathcal{A}_x} \|z\|. \end{aligned} \quad (69)$$

Here we used $c_x \epsilon N_0 \leq 1$.

For W_z , equations (50), (66), and (67) yield

$$\begin{aligned} \dot{W}_z &= -c_y \dot{\tau} e^{-c_y \tau} V_z + e^{-c_y \tau} (\nabla_z V_z f_z + \nabla_z V_z \dot{x}) \\ &\leq -\frac{c_y \sigma_1}{\epsilon} e^{-N_0 c_y \tau} \|z\|^2 + N_{zx} \ell_y \|x\|_{\mathcal{A}_x} \|z\| \\ &\quad + (N_{zz} \ell_y + \ell_x M_x) \|z\|^2. \end{aligned} \quad (70)$$

Since $W = \phi_\mu(W_x) + \phi_\nu(W_z)$, combining (69)–(70) with (68) gives

$$\dot{W} \leq - \left(\frac{\|x\|_{\mathcal{A}_x}}{\|z\|} \right)^\top Q_\epsilon \begin{pmatrix} \|x\|_{\mathcal{A}_x} \\ \|z\| \end{pmatrix}, \quad (71)$$

where

$$Q_\epsilon = \begin{pmatrix} Q_{11} & -Q_{12} \\ -Q_{12} & q_{22}/\epsilon - C_0 \end{pmatrix}$$

with $Q_{11} := \frac{1}{2} m_\mu a$, $Q_{12} := \frac{1}{2} (M_\mu e c N_x + M_\nu N_{zx} \ell_y)$, $q_{22} := m_\nu c_y \sigma_1 e^{-N_0 c_y \tau} \underline{b}$, and $C_0 := M_\nu (N_{zz} \ell_y + \ell_x M_x)$. Thus $Q_{11} > 0$, while the (2,2) entry grows as $1/\epsilon$. For example, it is sufficient to choose

$$0 < \epsilon \leq \epsilon_F^* := \min \left\{ 1, \frac{q_{22}}{2(C_0 + Q_{12}^2/Q_{11})} \right\}. \quad (72)$$

Then Q_ϵ is positive definite, uniformly on Ω_0^ϵ , and hence

$$\dot{W} < 0 \quad \text{on } \Omega_0^\epsilon \cap \mathcal{C} \setminus (\mathcal{A}_x \times \{0\} \times [0, N_0]). \quad (73)$$

Decrease across jumps

We next estimate the effect of the slow jump. Since $\mathcal{K}$ is fixed and compact and V_x, V_y are continuously differentiable on open neighborhoods of the corresponding compact sets, there exist $\ell_0 > 0$ and constants $L_x^J, L_y^J > 0$, independent of ϵ and $\ell_J \in [0, \ell_0]$, such that the following mean-value estimates hold for every jump from $\mathcal{K}$:

$$V_x(g_x(x, z + h(x))) - V_x(x) \leq L_x^J \|g_x(x, z + h(x)) - x\|, \quad (74)$$

$$\begin{aligned} V_y(g_y(x, z + h(x)), g_y(x, y)) - V_y(x, g_y(x, y)) &\leq \\ L_y^J \|g_y(x, z + h(x)) - x\|. \end{aligned} \quad (75)$$

Reducing ℓ_0 if necessary, the jump images of W_x, W_z also remain in the compact intervals on which (68) holds. This is possible because (63) provides a uniform positive distance from the poles of the two rational functions.

Using (54), (74), $e^{c_x \epsilon} - 1 \geq c_x \epsilon$, and $V_x \geq \underline{a} \|x\|_{\mathcal{A}_x}^2$, we obtain

$$\begin{aligned} W_x^+ - W_x &= e^{c_x \epsilon (\tau-1)} [V_x(g_x(x, z + h(x))) - e^{c_x \epsilon} V_x(x)] \\ &\leq -c_x \epsilon \underline{a} \|x\|_{\mathcal{A}_x}^2 + e L_x^J \ell_J \|z\|^2. \end{aligned} \quad (76)$$

For the fast part, Assumption 2 and (75) give

$$\begin{aligned} V_z(x^+, z^+) &= V_y(g_y(x, z + h(x)), g_y(x, y)) \\ &\leq V_y(x, g_y(x, y)) + L_y^J \ell_J \|z\|^2 \\ &\leq \gamma V_y(x, y) + L_y^J \ell_J \|z\|^2. \end{aligned} \quad (77)$$

Set $\lambda_J := 1 - e^{c_y \gamma} > 0$, and $\beta_J := \lambda_J e^{-N_0 c_y \tau} \underline{b} > 0$. Since $\tau^+ = \tau - 1$, it follows from (77) that

$$\begin{aligned} W_z^+ - W_z &\leq -\lambda_J W_z + e^{c_y \tau} L_y^J \ell_J \|z\|^2 \\ &\leq -\beta_J \|z\|^2 + e^{c_y \tau} L_y^J \ell_J \|z\|^2. \end{aligned} \quad (78)$$

Applying the mean-value theorem to the two scalar rational functions in (60), and using (68), (76), and (78), we obtain

$$\begin{aligned} W^+ - W &\leq -m_\mu c_x \epsilon \underline{a} \|x\|_{\mathcal{A}_x}^2 \\ &\quad - \left[m_\nu \beta_J - \ell_J (M_\mu e L_x^J + M_\nu e^{c_y \tau} L_y^J) \right] \|z\|^2. \end{aligned} \quad (79)$$

Therefore, after possibly reducing ℓ_0 as explained above, choose

$$\ell_J^* := \min \left\{ \ell_0, \frac{m_\nu \beta_J}{2(M_\mu e L_x^J + M_\nu e^{c_y \tau} L_y^J)} \right\}. \quad (80)$$

For every $0 \leq \ell_J \leq \ell_J^*$ and every $\epsilon \in (0, 1]$,

$$W^+ - W \leq -c_x^J \epsilon \|x\|_{\mathcal{A}_x}^2 - c_z^J \|z\|^2, \quad (81)$$

where $c_x^J := m_\mu c_x \underline{a} > 0$, and $c_z^J := \frac{1}{2} m_\nu \beta_J > 0$. In particular, every admissible jump strictly decreases W away from $\mathcal{A}_x \times \{0\} \times [0, N_0]$. The same conclusion holds during a burst of consecutive jumps, since $\tau^+ = \tau - 1$ and hence at most N_0 jumps can occur without intervening flow.

Conclusion

The construction above has the following order. The compact sets K_x^0 and K_y^0 are fixed first. They determine μ , $\hat{K}_x$, the Lyapunov function V_y , ν , and the fixed compact set $\mathcal{K}$. All constants appearing in the flow and jump estimates are then fixed on $\mathcal{K}$. Only after these choices do we select ℓ_J^* from (80) and ϵ^* from (72). In particular, neither the compact region on which the estimates are made nor any of its defining constants is chosen as a function of ϵ or ℓ_J .

Set $\epsilon^* := \epsilon_F^*$. Then, for every $0 \leq \ell_J \leq \ell_J^*$ and every $0 < \epsilon \leq \epsilon^*$, equations (73) and (81) show that W decreases along every admissible flow and every admissible jump in Ω_0^ϵ . Since the prescribed initial compact set is contained in the interior of Ω_0^ϵ by (62), the set Ω_0^ϵ is forward invariant. Thus every corresponding solution remains in the fixed compact set $\mathcal{K}$ and is bounded.

For completeness, the timer satisfies, between any two hybrid times (s, i) and (t, j) on the same solution,

$$\frac{\sigma_1}{\epsilon}(t - s) - N_0 \leq j - i \leq \frac{\sigma_2}{\epsilon}(t - s) + N_0, \quad (82)$$

which follows by integrating $\epsilon \dot{\tau} \in [\sigma_1, \sigma_2]$ and using $\tau^+ = \tau - 1$ with $\tau \in [0, N_0]$. Hence there are only finitely many jumps on every bounded interval of continuous time. Under the basic assumptions for (10), the bounded maximal solutions considered above are therefore complete.

The function W is positive definite with respect to $\mathcal{A} := \mathcal{A}_x \times \{0\} \times [0, N_0]$ relative to Ω_0^ϵ . The strict flow and jump inequalities above, together with the standard hybrid Lyapunov theorem (for example, [15, Theorem 3.18]), yield

$$\|x(t, j)\|_{\mathcal{A}_x} \rightarrow 0, \quad \|z(t, j)\| \rightarrow 0$$

as $t + j \rightarrow \infty$. By (82), continuous time also tends to infinity along complete solutions. Since $z = y - h(x)$, this proves (17)–(18).

Finally, K_x^0 and K_y^0 were arbitrary compact subsets of C_x and C_y . Hence the same construction applies to any prescribed compact set of initial conditions, with ℓ_J^* and ϵ^* chosen as functions of that fixed compact set. ■

APPENDIX II TECHNICAL DETAILS FOR SECTION V

We collect here the additional calculations and proof sketches needed for the stability analysis of Section V.

A. Projection of the QSS dynamics

Set $\tilde{\chi} = 0$. Then $\xi = 1_N \chi_o$, $\theta = \eta$, and (39) follows immediately from the compensated-coordinate flow. Write

$$\begin{aligned} \omega &= 1_N \omega_o + R \tilde{\omega}, & \bar{\psi} &:= \psi(\omega_o, \chi_o), \\ \Delta_\omega &:= \Psi_c - I_N \otimes \bar{\psi}. \end{aligned} \quad (83)$$

Since $\psi_o(\omega_i, \chi_o) = -\omega_i + \chi_o$,

$$\Psi_{o,c} = 1_N \psi_o(\omega_o, \chi_o) - R \tilde{\omega}.$$

Using $\eta = (1_N \otimes I_2) \eta_o + (R \otimes I_2) \tilde{\eta}$ therefore yields

$$\begin{aligned} q_c &= \Psi_c \eta + \Psi_{o,c} \\ &= (I_N \otimes \bar{\psi}) [(1_N \otimes I_2) \eta_o + (R \otimes I_2) \tilde{\eta}] \\ &\quad + \Delta_\omega [(1_N \otimes I_2) \eta_o + (R \otimes I_2) \tilde{\eta}] + 1_N \psi_o(\omega_o, \chi_o) - R \tilde{\omega}. \end{aligned}$$

Using $(I_N \otimes \bar{\psi})(1_N \otimes I_2) \eta_o = 1_N \bar{\psi} \eta_o$, and $(I_N \otimes \bar{\psi})(R \otimes I_2) \tilde{\eta} = (R \otimes \bar{\psi}) \tilde{\eta}$, and letting

$$\delta_q := \Delta_\omega [(1_N \otimes I_2) \eta_o + (R \otimes I_2) \tilde{\eta}] - R \tilde{\omega}, \quad (84)$$

we get

$$q_c = 1_N [\bar{\psi} \eta_o + \psi_o(\omega_o, \chi_o)] + (R \otimes \bar{\psi}) \tilde{\eta} + \delta_q. \quad (85)$$

To find a bound on the norm of δ_q , we note that the i th diagonal block of Δ_ω is

$$\psi(\omega_i, \chi_o) - \psi(\omega_o, \chi_o).$$

Since ψ is continuously differentiable, on the fixed compact region there is $L_\psi > 0$ such that

$$\|\psi(\omega_i, \chi_o) - \psi(\omega_o, \chi_o)\| \leq L_\psi |\omega_i - \omega_o|.$$

Hence, using $\omega - 1_N \omega_o = R \tilde{\omega}$ and $\|R\| = 1$,

$$\|\Delta_\omega\| \leq L_\psi \|\tilde{\omega}\|. \quad (86)$$

It follows from (84) that

$$\begin{aligned} \|\delta_q\| &\leq \|\Delta_\omega\| (\|(1_N \otimes I_2) \eta_o\| + \|(R \otimes I_2) \tilde{\eta}\|) + \|R \tilde{\omega}\| \\ &\leq \left[1 + L_\psi (\sqrt{N} \|\eta_o\| + \|\tilde{\eta}\|)\right] \|\tilde{\omega}\|. \end{aligned}$$

Since η_o and $\tilde{\eta}$ are bounded on the fixed compact region, there exists $c_q > 0$ such that

$$\|\delta_q\| \leq c_q \|\tilde{\omega}\|. \quad (87)$$

The compact region and L_ψ can be chosen uniformly for $0 < \alpha \leq \alpha_0$, and therefore so can c_q .

For the derivation of (41), we note that the first equation of (39) gives $\dot{\omega}_o = -\omega_o + \chi_o$ and $\dot{\tilde{\omega}} = -\tilde{\omega}$. Furthermore,

$$P_R q_c = (R \otimes \bar{\psi}) \tilde{\eta} + P_R \delta_q, \quad \Psi_c^\top = I_N \otimes \bar{\psi}^\top + \Delta_\omega^\top.$$

Projecting $\dot{\eta} = -\alpha \Psi_c^\top P_R q_c$ onto the average and disagreement coordinates gives (41), with

$$d_o := \frac{1}{N} 1_N^\top \delta_q, \quad (88)$$

$$d_{\eta_o} := -\frac{\alpha}{N} (1_N^\top \otimes I_2) \Delta_\omega^\top [(R \otimes \bar{\psi}) \tilde{\eta} + P_R \delta_q], \quad (89)$$

$$d_{\eta_e} := -\alpha (R^\top \otimes I_2) \{\Delta_\omega^\top (R \otimes \bar{\psi}) \tilde{\eta} + \Psi_c^\top P_R \delta_q\}. \quad (90)$$

The terms containing $I_N \otimes \bar{\psi}^\top$ disappear from d_{η_o} because $1_N^\top R = 1_N^\top P_R = 0$. Equations (86)–(87), together with compactness, give

$$\begin{aligned} |d_o| &\leq c_d \|\tilde{\omega}\|, \\ \|d_{\eta_o}\| &\leq c_d \alpha \|\tilde{\omega}\| (\|\tilde{\eta}\| + \|\tilde{\omega}\|), \\ \|d_{\eta_e}\| &\leq c_d \alpha \|\tilde{\omega}\|. \end{aligned} \quad (91)$$

The weaker bound (42) follows on the fixed compact region. Notice that all perturbations vanish identically on $\tilde{\omega} = 0$.

B. Proof of Proposition 1

Put $u = s_\omega$. Since $\dot{u} = -u + s_o$, one has $s_o = u + \dot{u}$. Substitution into (43b), using the definitions of ψ and ψ_o , gives

$$\ddot{u} + \mu_o(u^2 - 1)\dot{u} + \nu_o u = 0. \quad (92)$$

With $r = \sqrt{\nu_o} t$, (92) becomes the standard Van der Pol equation

$$\frac{d^2 u}{dr^2} + \lambda_o(u^2 - 1) \frac{du}{dr} + u = 0, \quad \lambda_o := \frac{\mu_o}{\sqrt{\nu_o}} > 0.$$

The classical Liénard theorem gives existence, uniqueness, and orbital stability of its nontrivial periodic orbit; see [17, Sec. 3.8, pp. 253–257]. Hyperbolicity and stability of the Van der Pol cycle are stated explicitly in [18, Prop. 2.2, p. 5]. Since the periodic orbit is hyperbolic, it persists smoothly under small variations of θ_o ; after fixing the phase by a local transversal section, both the periodic solution and its period depend smoothly on θ_o . Compactness of Θ_o^* yields uniform local exponential orbital stability and uniform upper and lower bounds on the period.

It remains to establish uniform excitation estimate in (44). Along a periodic orbit, $-s_\omega + s_o = \dot{s}_\omega$, and hence

$$\psi(t) = \psi(s_\omega(t), s_o(t)) = [-(s_\omega^2 - 1)\dot{s}_\omega \quad -s_\omega]^\top. \quad (93)$$

Suppose that $v = \text{col}(v_1, v_2)$ satisfies $\psi(t)v \equiv 0$ over one period $[0, T]$. Multiplying by s_ω and integrating gives

$$-v_1 \int_0^T s_\omega(s_\omega^2 - 1)\dot{s}_\omega dt - v_2 \int_0^T s_\omega^2 dt = 0.$$

The first integral is

$$\left[\frac{s_\omega^4}{4} - \frac{s_\omega^2}{2} \right]_0^T = 0,$$

whereas the second is strictly positive on a nontrivial orbit. Hence $v_2 = 0$. Then $v_1(s_\omega^2 - 1)\dot{s}_\omega \equiv 0$. A nonconstant periodic orbit contains an open set on which $\dot{s}_\omega \neq 0$, and s_ω cannot equal ± 1 on such an interval; consequently $v_1 = 0$. Thus the one-period Gramian

$$G(\theta_o) := \int_0^{T(\theta_o)} \psi^\top \psi dt$$

is positive definite. Its continuous dependence on θ_o and compactness of Θ_o^* imply

$$\underline{\mu} := \min_{\theta_o \in \Theta_o^*} \lambda_{\min} G(\theta_o) > 0.$$

Taking $T_{\text{PE}} := \max_{\theta_o \in \Theta_o^*} T(\theta_o)$ proves the lower bound in (44): every interval of length T_{PE} contains a complete period, and the remaining Gramian contribution is positive semidefinite.

For the upper bound in (44), we note that the family of periodic orbits is compact and ψ is continuous, so there exists $M_{\psi, \text{cyc}} > 0$ such that $\|\psi\| \leq M_{\psi, \text{cyc}}$ uniformly on all $\mathcal{A}_s(\theta_o)$, $\theta_o \in \Theta_o^*$. Hence

$$\int_t^{t+T_{\text{PE}}} \psi^\top \psi ds \preceq T_{\text{PE}} M_{\psi, \text{cyc}}^2 I_2,$$

so that the upper bound in (44) with $\bar{\mu} = T_{\text{PE}} M_{\psi, \text{cyc}}^2$ holds.

To conclude this section, we note that $(\theta_o, \varphi) \mapsto s^{\theta_o}(\varphi T(\theta_o))$, $\varphi \in [0, 1]$, is C^1 on $\Theta_o^* \times [0, 1]$; by compactness its θ_o -derivative is bounded, so there is $L_{\mathcal{A}} > 0$ with

$$d_H(\mathcal{A}_s(\theta_o), \mathcal{A}_s(\theta'_o)) \leq L_{\mathcal{A}} |\theta_o - \theta'_o| \quad (94)$$

for $\theta_o, \theta'_o \in \Theta_o^*$. Here, d_H denotes Hausdorff distance.

C. Proof sketch of Proposition 2

Denote the prescribed initial set by $\mathcal{S}$. Fix the data of the proposition and put $s := \text{col}(\omega_o, \chi_o)$. The construction separates oscillator disagreement, orbital attraction, and parameter disagreement. First, (41) gives

$$V_\omega(\tilde{\omega}) := \|\tilde{\omega}\|^2, \quad \dot{V}_\omega = -2\|\tilde{\omega}\|^2. \quad (95)$$

For $\tilde{\omega} = 0$, the variables (s, η_o) satisfy (43) with η_o constant. Uniform exponential orbital stability supplies a smooth local Lyapunov function $V_s(s, \eta_o)$, with constants uniform on the compact parameter family, such that

$$c_s \|s\|_{\mathcal{A}_s(\eta_o)}^2 \leq V_s(s, \eta_o) \leq \bar{c}_s \|s\|_{\mathcal{A}_s(\eta_o)}^2, \quad (96)$$

$$\|\nabla_{(s, \eta_o)} V_s(s, \eta_o)\| \leq c_s \|s\|_{\mathcal{A}_s(\eta_o)}, \quad (97)$$

$$\dot{V}_s(s, \eta_o) \leq -a_s \|s\|_{\mathcal{A}_s(\eta_o)}^2. \quad (98)$$

The full gradient vanishes on the cycle family, yielding (97) by smoothness on a compact tube.

For each block $\tilde{\eta}_i \in \mathbb{R}^2$, the remaining reduced dynamics are

$$\dot{\tilde{\eta}}_i = -\alpha \bar{\psi}^\top \bar{\psi} \tilde{\eta}_i, \quad \bar{\psi} := \psi(s). \quad (99)$$

Let $\bar{\psi}(r; s, \eta_o)$ denote the regressor evaluated along (43), starting from s with parameter η_o , and put $T := T_{\text{PE}}$. Define

$$H(s, \eta_o) := \int_0^T (T - r) \bar{\psi}(r; s, \eta_o)^\top \bar{\psi}(r; s, \eta_o) dr. \quad (100)$$

The PE estimate persists on a sufficiently small tube, so

$$S(s, \eta_o) := \int_0^T \bar{\psi}(r; s, \eta_o)^\top \bar{\psi}(r; s, \eta_o) dr \succeq \mu_{\text{PE}} I_2 \quad (101)$$

for some $\mu_{\text{PE}} > 0$. Differentiation along (43) gives

$$\dot{H}(s, \eta_o) = S(s, \eta_o) - T \bar{\psi}^\top \bar{\psi}. \quad (102)$$

Choose $\delta > 0$ with $\delta T \leq 1$ and set

$$\begin{aligned} V_{\eta, i}(\tilde{\eta}_i, s, \eta_o) &:= \tilde{\eta}_i^\top [I_2 - \delta \alpha H(s, \eta_o)] \tilde{\eta}_i, \\ V_{\tilde{\eta}}(\tilde{\eta}, s, \eta_o) &:= \sum_{i=1}^{N-1} V_{\eta, i}(\tilde{\eta}_i, s, \eta_o). \end{aligned} \quad (103)$$

For sufficiently small α , these forms are uniformly positive definite. Writing $H = H(s, \eta_o)$ and $S = S(s, \eta_o)$, differentiation gives

$$\begin{aligned}\dot{V}_{\eta,i} &= -\delta\alpha\tilde{\eta}_i^\top S\tilde{\eta}_i - \alpha(2 - \delta T)\|\tilde{\psi}\tilde{\eta}_i\|^2 \\ &\quad + \delta\alpha^2\tilde{\eta}_i^\top (\tilde{\psi}^\top \tilde{\psi} H + H\tilde{\psi}^\top \tilde{\psi})\tilde{\eta}_i \\ &\leq -a_\eta\alpha\|\tilde{\eta}_i\|^2,\end{aligned}\quad (104)$$

where $a_\eta > 0$ is independent of α . In particular,

$$\begin{aligned}\frac{1}{2}\|\tilde{\eta}\|^2 &\leq V_{\tilde{\eta}}(\tilde{\eta}, s, \eta_o) \leq \frac{3}{2}\|\tilde{\eta}\|^2, \\ \dot{V}_{\tilde{\eta}} &\leq -a_\eta\alpha\|\tilde{\eta}\|^2.\end{aligned}\quad (105)$$

Thus PE produces strict parameter-error decay of order α , without degenerating the quadratic bounds as $\alpha \downarrow 0$.

To cover the prescribed compact basin family, the local certificates must be extended along solutions of (43), with η_o held constant and integral corrections preserving the required derivative estimates. The extension and smooth gluing, and the construction of C_x , its uniform interior margin, and the level $\bar{v}$, are omitted here and must be supplied in a complete proof. Write $V_s^{\text{ext}}(s, \eta_o)$ and $H^{\text{ext}}(s, \eta_o)$ for such extensions, and use H^{ext} in (103). The composite function is

$$V_x^\alpha(x) := \rho_s V_s^{\text{ext}}(s, \eta_o) + V_{\tilde{\eta}}(\tilde{\eta}, s, \eta_o) + \rho_\omega V_\omega(\tilde{\omega}), \quad (106)$$

with positive weights independent of α . On the fixed compact region, (91) bounds the additional terms by

$$C\|s\|_{\mathcal{A}_s(\eta_o)}\|\tilde{\omega}\| + C\alpha\|\tilde{\eta}\|\|\tilde{\omega}\|.$$

Young's inequality, a sufficiently large ρ_ω , and a smaller α_0 give

$$\dot{V}_x^\alpha \leq -a'_s\|s\|_{\mathcal{A}_s(\eta_o)}^2 - a'_\eta\alpha\|\tilde{\eta}\|^2 - a'_\omega\|\tilde{\omega}\|^2. \quad (107)$$

The coordinate decomposition and (94) show that $\|x\|_{\mathcal{A}_x}^2$ lies between positive constant multiples of

$$\|s\|_{\mathcal{A}_s(\eta_o)}^2 + \|\tilde{\omega}\|^2 + \|\tilde{\eta}\|^2.$$

Consequently, the composite quadratic, decay, and full-gradient bounds have the form (46), uniformly in α . A complete regional construction must also retain a uniform absolute gradient bound on a fixed neighborhood of C_x for the subsequent jump estimates. The QSS dynamics contain no ϵ .

D. Proof of Proposition 3

For fixed x , the fast jump dynamics are $y^+ = \Lambda y$. The gain condition makes Λ symmetric Schur, so $V_y(x, y) := \|y\|^2$ satisfies

$$V_y(x, y^+) \leq \gamma V_y(x, y), \quad \gamma := \|\Lambda\|^2 < 1.$$

Moreover, $\nabla_y V_y = 2y$ and $\nabla_x V_y = 0$. Thus Assumption 2 holds with equilibrium map $h \equiv 0$.

For Assumption 3, note that $f_y(x, 0, \epsilon) = R^\top q_c$. On $\mathcal{A}_x$, one has $\tilde{\omega} = \tilde{\eta} = 0$, and (85) gives

$$q_c = 1_N[\psi(s_\omega, s_o)\eta_o + \psi_o(s_\omega, s_o)].$$

Hence $R^\top q_c = 0$, which is the required invariance condition.

Finally, fix a compact region containing the states under consideration and the interpolation segments between their output jump values. Let L_Ψ bound $D_\xi \Psi$ on this region. The compensated-coordinate jump map gives

$$\|\eta^+ - \eta\| \leq \alpha\|\Psi(\omega, \xi^+) - \Psi(\omega, \xi)\| \|R\Lambda y\|.$$

Since $\xi^+ - \xi = -\kappa R\Lambda \mathcal{L}y$ and $\|R\| = 1$,

$$\|\eta^+ - \eta\| \leq c_\eta \alpha \kappa \|y\|^2, \quad (108)$$

where $c_\eta > 0$ may be chosen independently of α and ϵ on the fixed region. The variables ω and χ_o do not jump, so

$$\|x^+ - x\| = \|\eta^+ - \eta\| \leq c_\eta \alpha \kappa \|y\|^2.$$

Thus Assumption 4 holds with $\ell_J = c_\eta \alpha \kappa$.

E. Proof sketch of Theorem 2

We apply the estimates of Theorem 1 after showing that the slow state x reaches the initial set specified in Proposition 2, contained in the region C_x where the required Lyapunov estimates hold. This initial set is constructed from $\mathcal{K}_0$ in the proof. Solutions of (43) with constant η_o provide a reference evolution leading into this set. Small initial $\|\tilde{\omega}\|$, together with sufficiently small α and ϵ , keeps the actual trajectory sufficiently close to this reference to reach the set in finite time while remaining bounded. At the resulting point (T_0, j_0) , the finite-time boundedness estimates provide a constant $R_\chi > 0$ such that

$$\|y(T_0, j_0)\| = \|\tilde{\chi}(T_0, j_0)\| \leq R_\chi,$$

uniformly over the prescribed initial conditions and sufficiently small α, ϵ . We therefore apply the estimates of Theorem 1 from this point, using $K_y^0 := \{y : \|y\| \leq R_\chi\}$ as the compact initial set for y , up to a possible exit from the prescribed region. The details establishing these finite-time estimates are omitted here.

Once x has entered the prescribed initial set inside C_x and $\|y\| = \|\tilde{\chi}\| \leq R_\chi$, we restart the argument of Theorem 1. Proposition 2 supplies the Lyapunov inequalities required by Assumption 1 on C_x . Proposition 3 verifies Assumptions 2–4: $V_y(y) = \|y\|^2$ decreases at jumps because $\|\Lambda\| < 1$, $R^\top q = 0$ on $\mathcal{A}_x \times \{0\}$, and

$$\|x^+ - x\| \leq \ell_J \|y\|^2, \quad \ell_J = c_\eta \alpha \kappa.$$

With $\nu := R_\chi^2 + 1$, the regional estimates use the bound $\|y\| \leq r_y$, where $r_y := [e^{N_0}(\nu + 1)]^{1/2}$. The resulting compact neighborhood and the uniform gradient bound determine c_η and a jump threshold $\ell_J^* > 0$ independently of α . We may therefore choose

$$0 < \alpha \leq \min\{\alpha_0, \ell_J^*/(c_\eta \kappa)\}, \quad 0 < \epsilon \leq \epsilon_{\text{sp}}(\alpha),$$

where $\epsilon_{\text{sp}}(\alpha) > 0$ is supplied by the proof of Theorem 1. At this stage, its estimates hold only until a possible exit from the prescribed region.

To make these estimates valid for all subsequent times, we must show that x remains in C_x and $\|y\| \leq r_y$. In particular,

the average parameter η_o must stay away from the boundary of Θ_o^* . The key estimates are

$$\begin{aligned} \|\dot{\eta}_o\| &\leq C\alpha(\|x\|_{\mathcal{A}_x} + \|y\|)(\|\tilde{\omega}\| + \|y\|), \\ \|\eta_o^+ - \eta_o\| &\leq c_\eta\alpha\kappa\|y\|^2. \end{aligned} \quad (109)$$

The required argument combines the regional Lyapunov estimates with these bounds to control the accumulated parameter change during flows and jumps. One must show that η_o has finite total variation, small enough to keep it in a fixed compact subset of $\text{int } \Theta_o^*$. Together with the uniform sublevel margin of Proposition 2, this excludes an exit from C_x . The estimates establishing this confinement, including the treatment of jumps, are omitted and need to be formalized.

Once entrance and confinement are established, choose the final thresholds α^* and $\epsilon^*(\alpha)$ to satisfy those requirements and the preceding restrictions from Theorem 1. Boundedness and the timer dynamics then ensure completeness, and Theorem 1 gives

$$\|x(t, j)\|_{\mathcal{A}_x} \rightarrow 0, \quad \|\tilde{\chi}(t, j)\| \rightarrow 0. \quad (110)$$

Finite total variation implies $\eta_o \rightarrow \theta_\infty \in (0, \infty)^2$. The coordinate decompositions and $\|\theta - \eta\| \leq \alpha\|\Psi\|\|\tilde{\chi}\| \rightarrow 0$ then yield state synchronization and $\theta_i \rightarrow \theta_\infty$. Finally, $\|s\|_{\mathcal{A}_s(\eta_o)} \rightarrow 0$ and (94) imply convergence of each (ω_i, ξ_i) to $\mathcal{A}_s(\theta_\infty)$, giving (49).